Comparative Analysis of LSTM-Attention and Temporal Fusion Transformer for Stock Forecasting

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Outline

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2. Dataset & EDA
3. Feature Engineering
4. Model Architectures
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7. Conclusion

Problem Statement

- Stock price forecasting is a challenging time series problem due to high volatility and non-linearity
- Traditional models like ARIMA fail to capture complex temporal dependencies
- Deep learning models — especially LSTM and Transformer-based architectures — have shown promising results
- **Goal:** Compare LSTM-Attention, TFT, and a Hybrid Ensemble on two contrasting datasets

Why These Two Stocks?

Dataset Choice — Systematic vs. Idiosyncratic Noise

- **TSLA (Tesla)** — Highly volatile individual stock, news-driven, unpredictable peaks
- **S&P 500 Index** — Diversified market index, smoother trends, less volatile

Hypothesis

Models will perform better on S&P 500 due to lower volatility.
Volatility-aware features like Bollinger Band Width will be more important for TSLA.

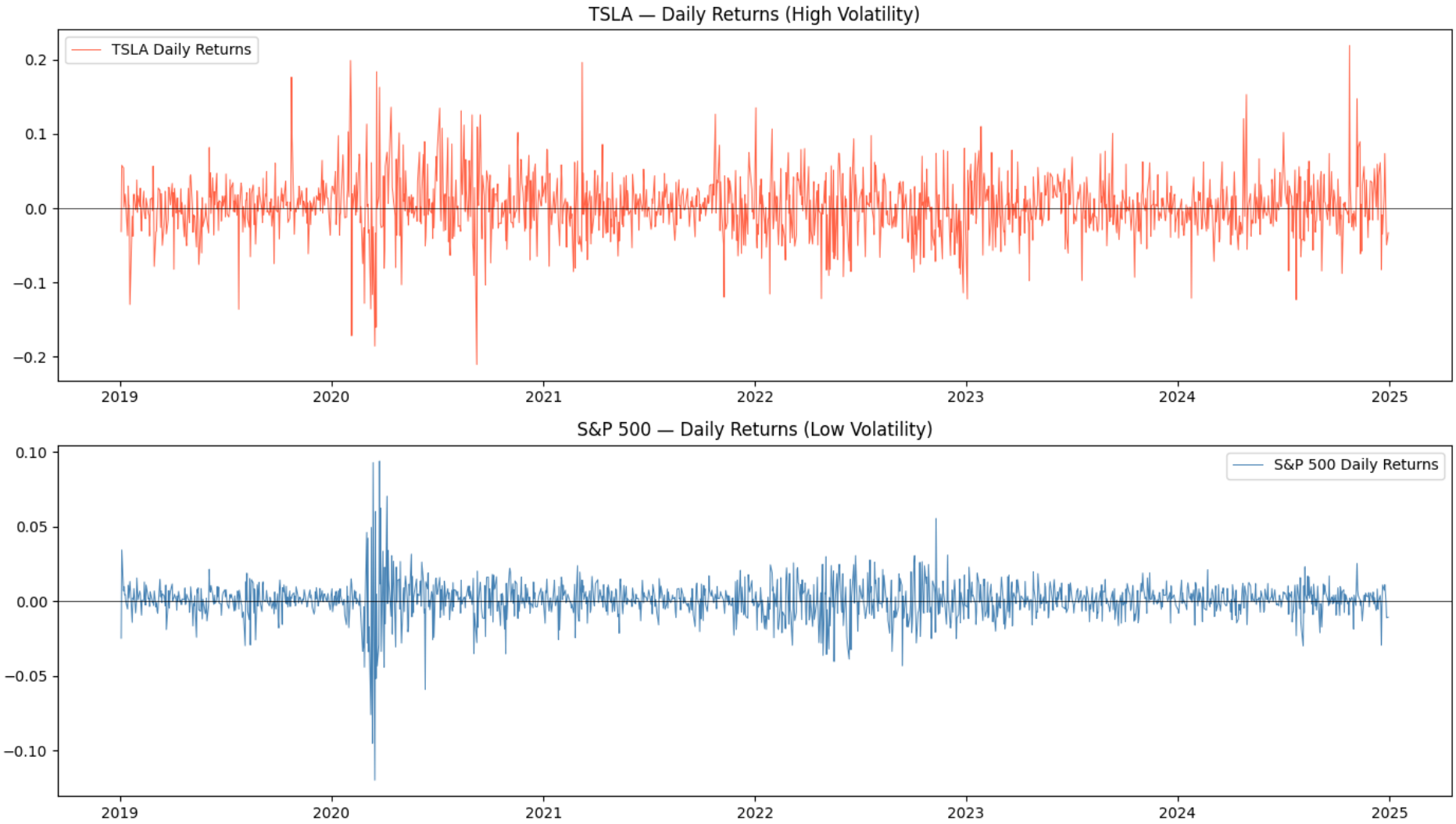
Data Collection

- Source: `yfinance` library
- Period: January 2019 — December 2024
- Features: OHLCV (Open, High, Low, Close, Volume)
- Train / Val / Test Split: **70% / 15% / 15%** (time-based, never random)
- Both datasets: **1460 trading days**

Why Time-Based Split?

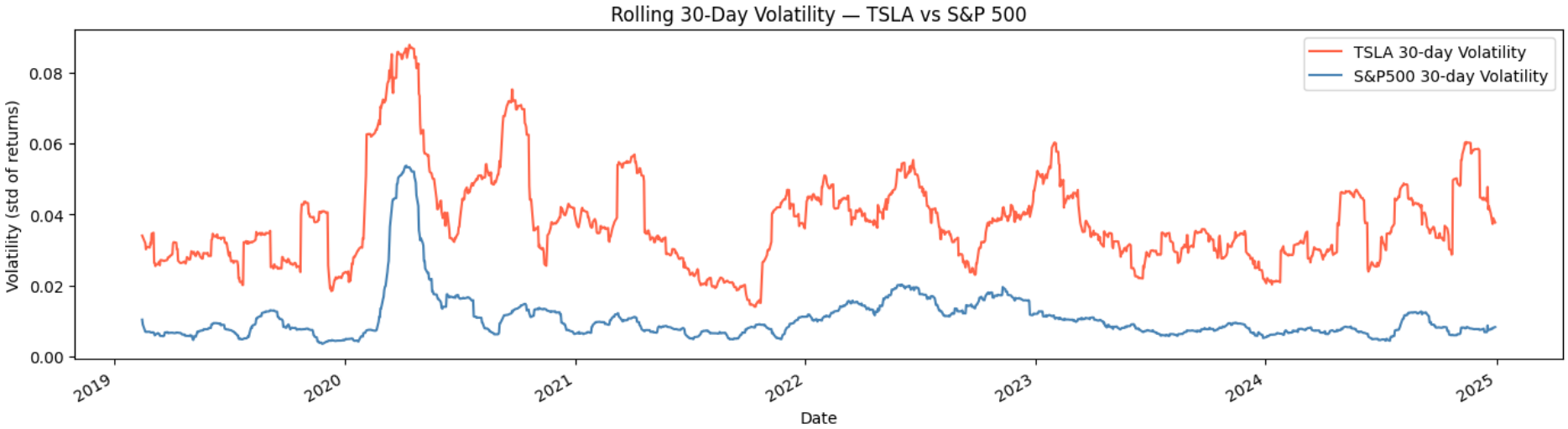
Random splitting would cause data leakage — future prices leaking into training data, giving falsely optimistic results.

EDA — Price & Volatility



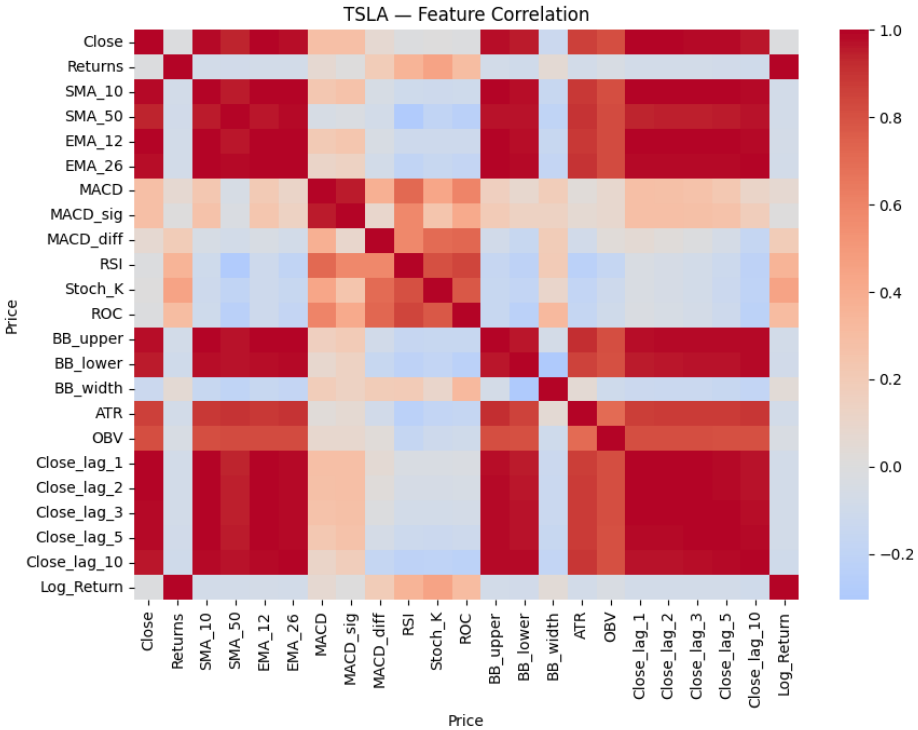
TSLA shows significantly higher price swings compared to the smoother S&P 500 trend.

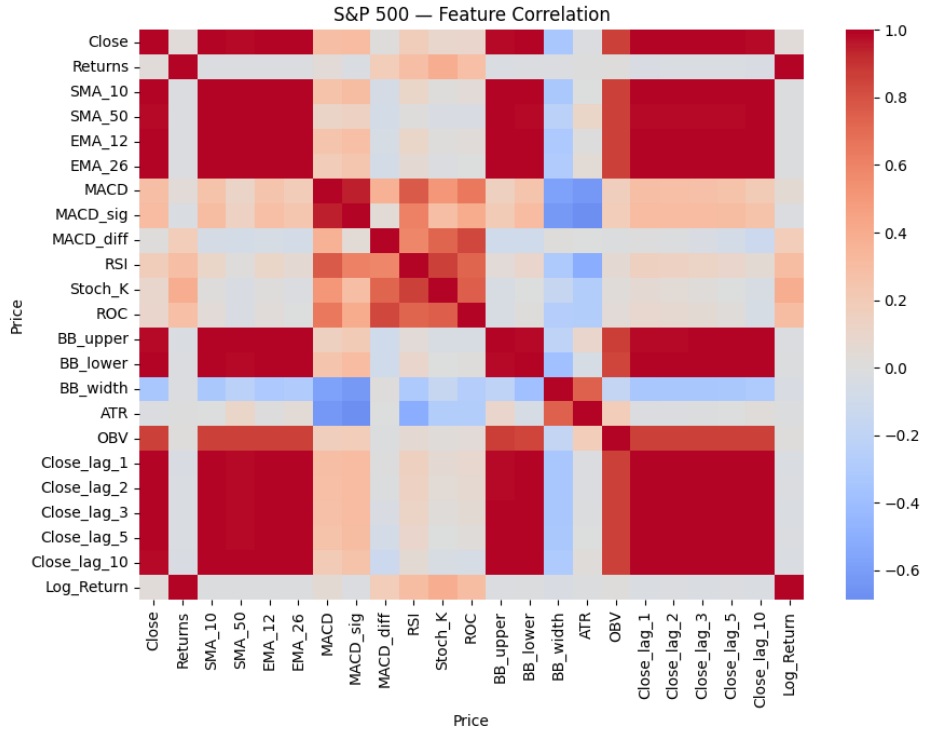
Volatility Comparison



- TSLA is consistently 3–4x more volatile than S&P 500
- This justifies using TSLA as a stress test for our models

Feature Correlation





Technical Indicators

Trend

- SMA (10, 50)
- EMA (12, 26)
- MACD, MACD Signal, MACD Diff

Momentum

- RSI (14)
- Stochastic Oscillator
- Rate of Change (ROC)

Volatility

- Bollinger Bands (Upper, Lower, Width)
- ATR (Average True Range)

Volume

- OBV (On-Balance Volume)

Lag Features

- Close lag 1, 2, 3, 5, 10
- Log Returns

Total: 23 features — same function applied to both datasets ensuring identical INPUT_SIZE

Why LSTM + Attention?

Standard LSTM Limitation

In long sequences (60 days), early information can be "forgotten" by the time the model reaches recent timesteps.

Attention Mechanism

Attention allows the model to look back at all 60 days and learn which specific timesteps matter most for predicting the next 5 days.

- Input: 60-day window of 23 features
- Architecture: LSTM (3 layers, 256 hidden) $\rightarrow$ Attention $\rightarrow$ FC (5 outputs)
- Output: Next 5 days of closing price
- Same architecture trained separately on TSLA and S&P 500

Why Temporal Fusion Transformer?

- TFT is specifically designed for **multi-horizon time series forecasting**
- Key components:
  - **Variable Selection Networks** — automatically selects important features
  - **Gated Residual Networks** — filters irrelevant information
  - **Multi-head Self-Attention** — captures long-range dependencies
  - **Quantile outputs** — provides prediction uncertainty
- Same TFT architecture used; hyperparameters were tuned for improved convergence

Hybrid Ensemble

Motivation

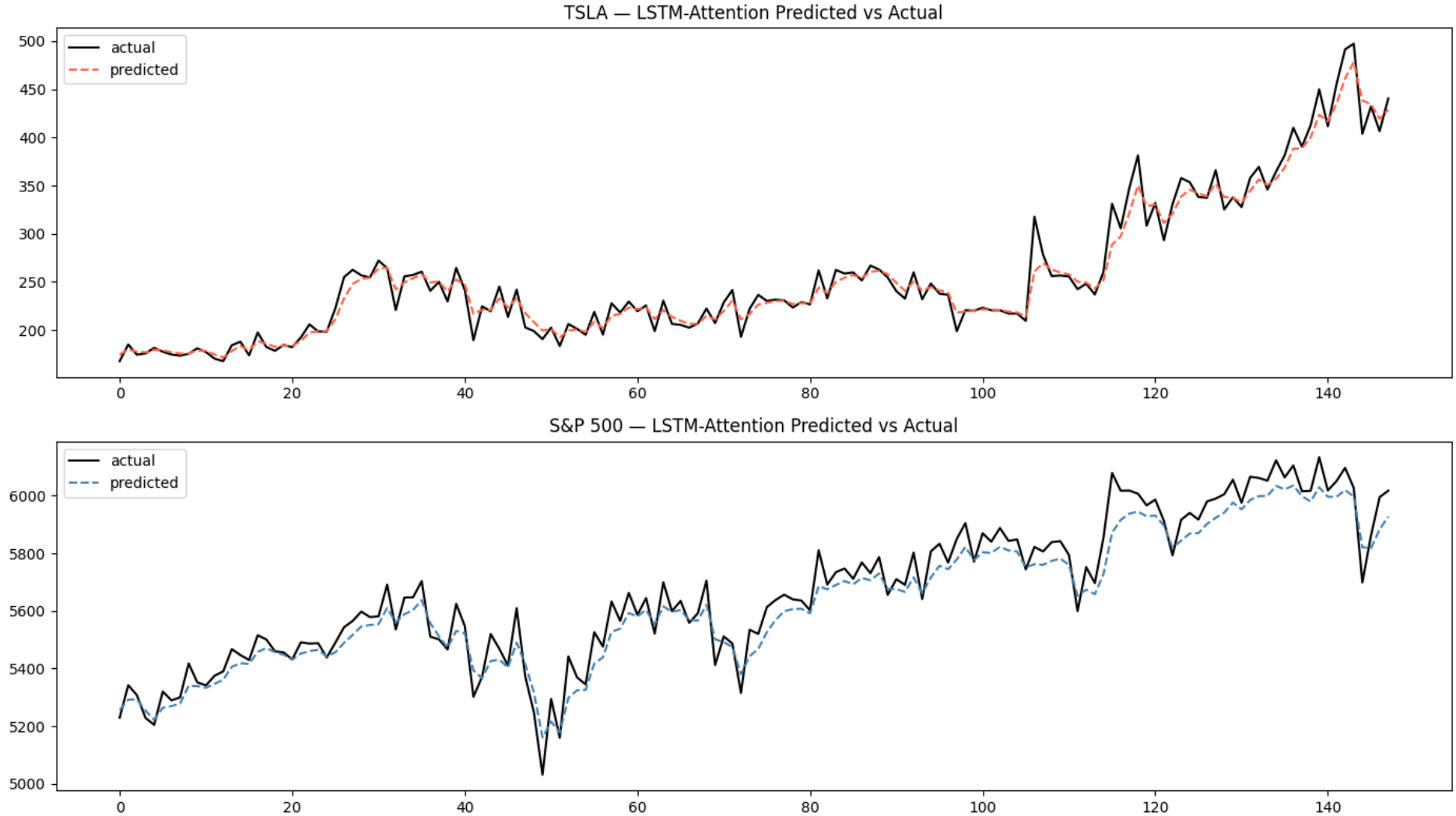
Individual models have complementary strengths:

- LSTM-Attention captures short-term sequential patterns
- TFT captures long-range dependencies with variable selection

$$\hat{y}_{ensemble} = w_{LSTM} \cdot \hat{y}_{LSTM} + w_{TFT} \cdot \hat{y}_{TFT} \tag{1}$$

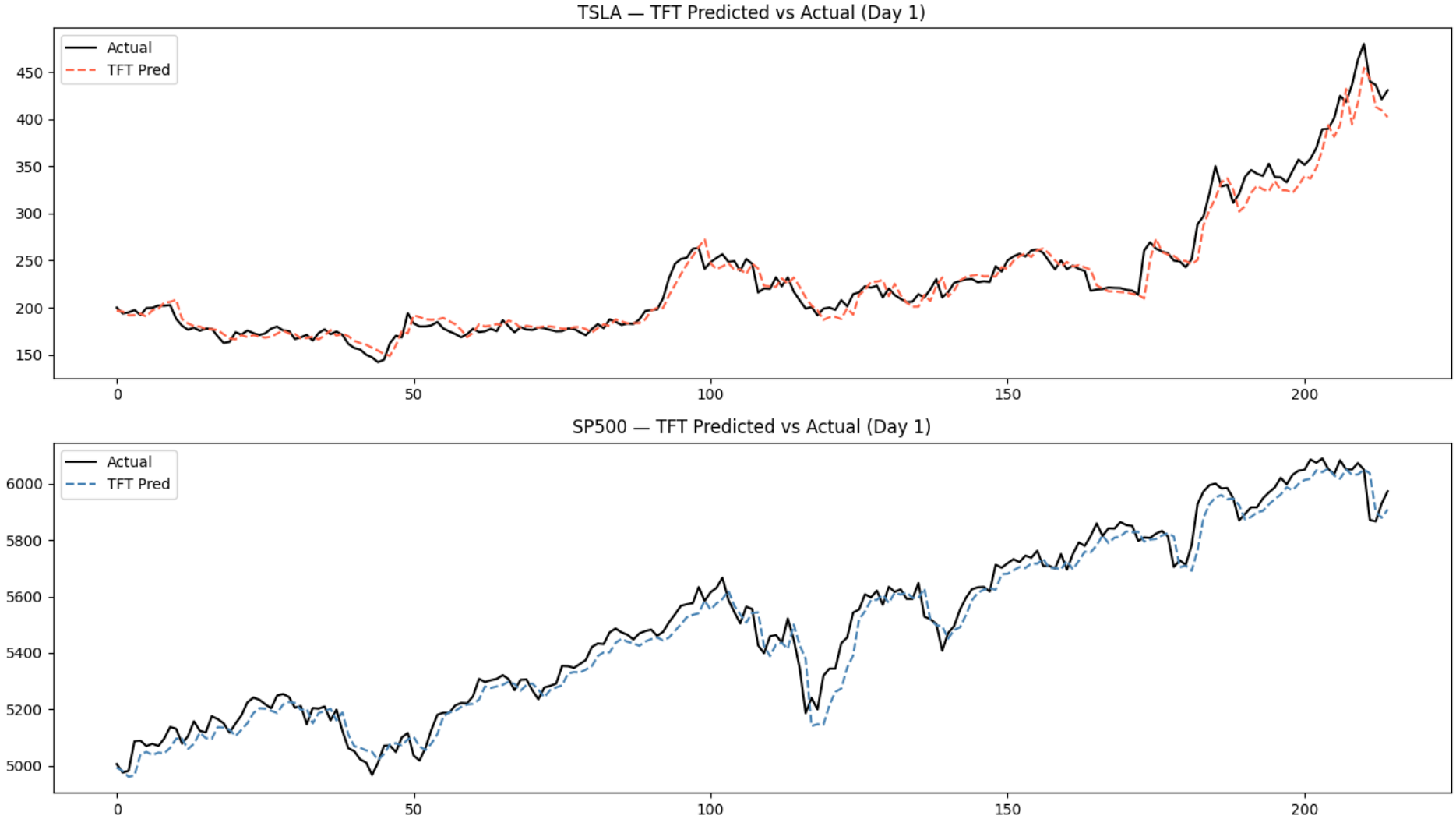
- Optimal weights found by minimizing RMSE on validation set
- TSLA: $w_{LSTM} = 0.3$, $w_{TFT} = 0.7$
- S&P 500: $w_{LSTM} = 0.3$, $w_{TFT} = 0.7$

LSTM-Attention Predictions



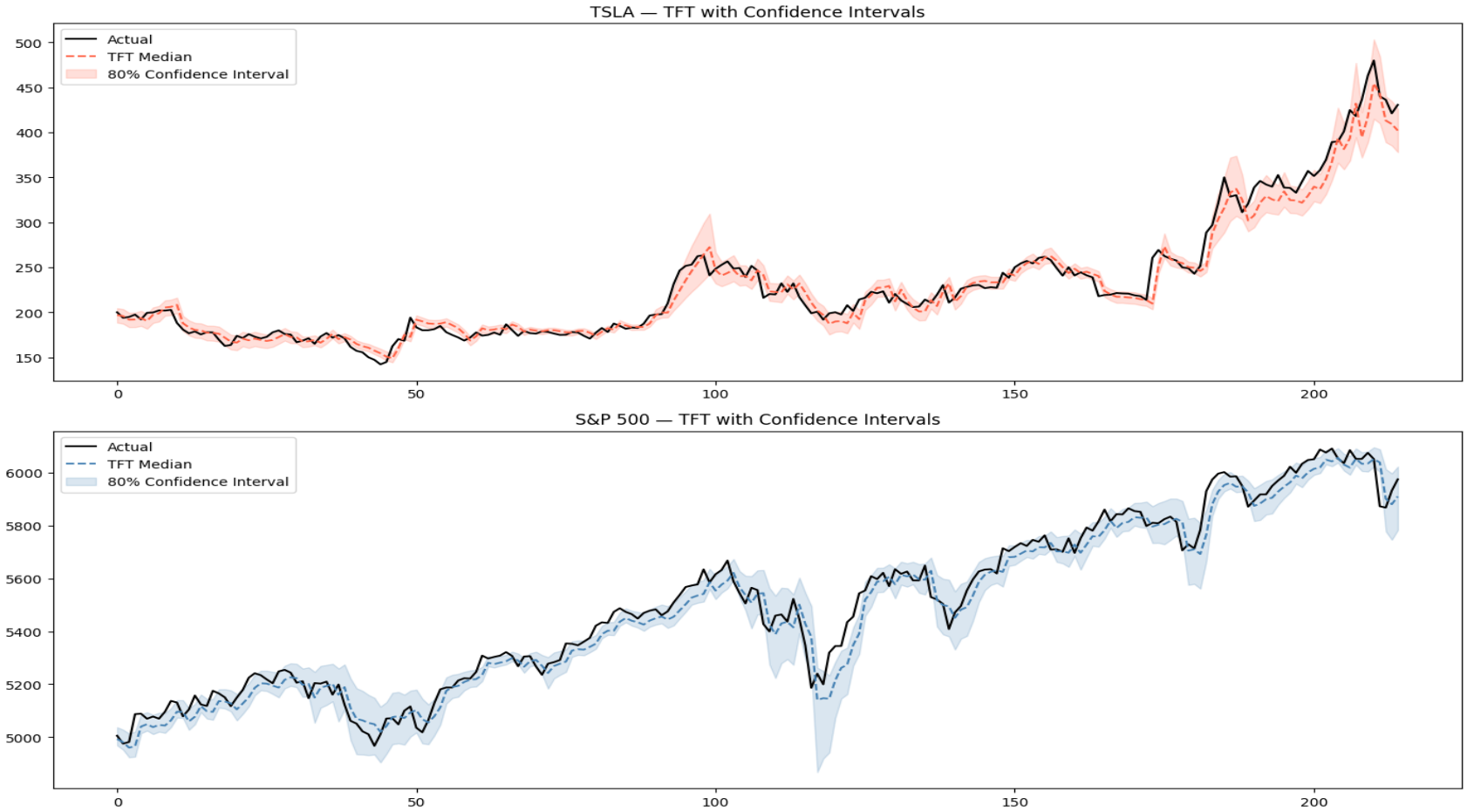
LSTM-Attention captures general trends well but shows more prediction error during sudden price spikes.

TFT Predictions



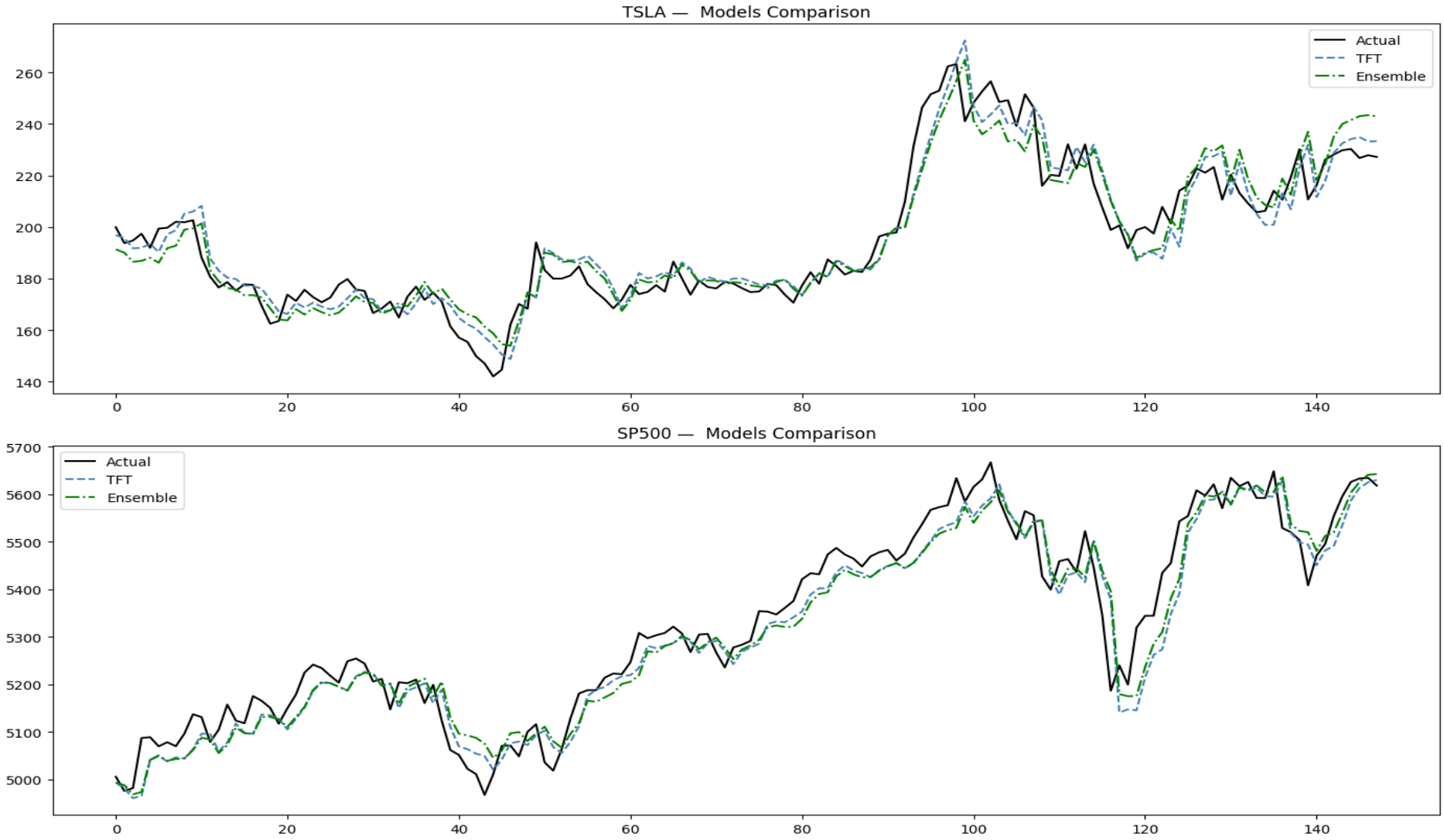
Hyperparameters were tuned to improve convergence stability,TFT follows actual prices more closely, especially for S&P 500 where trends are smoother.

TFT Forecast with Confidence Intervals



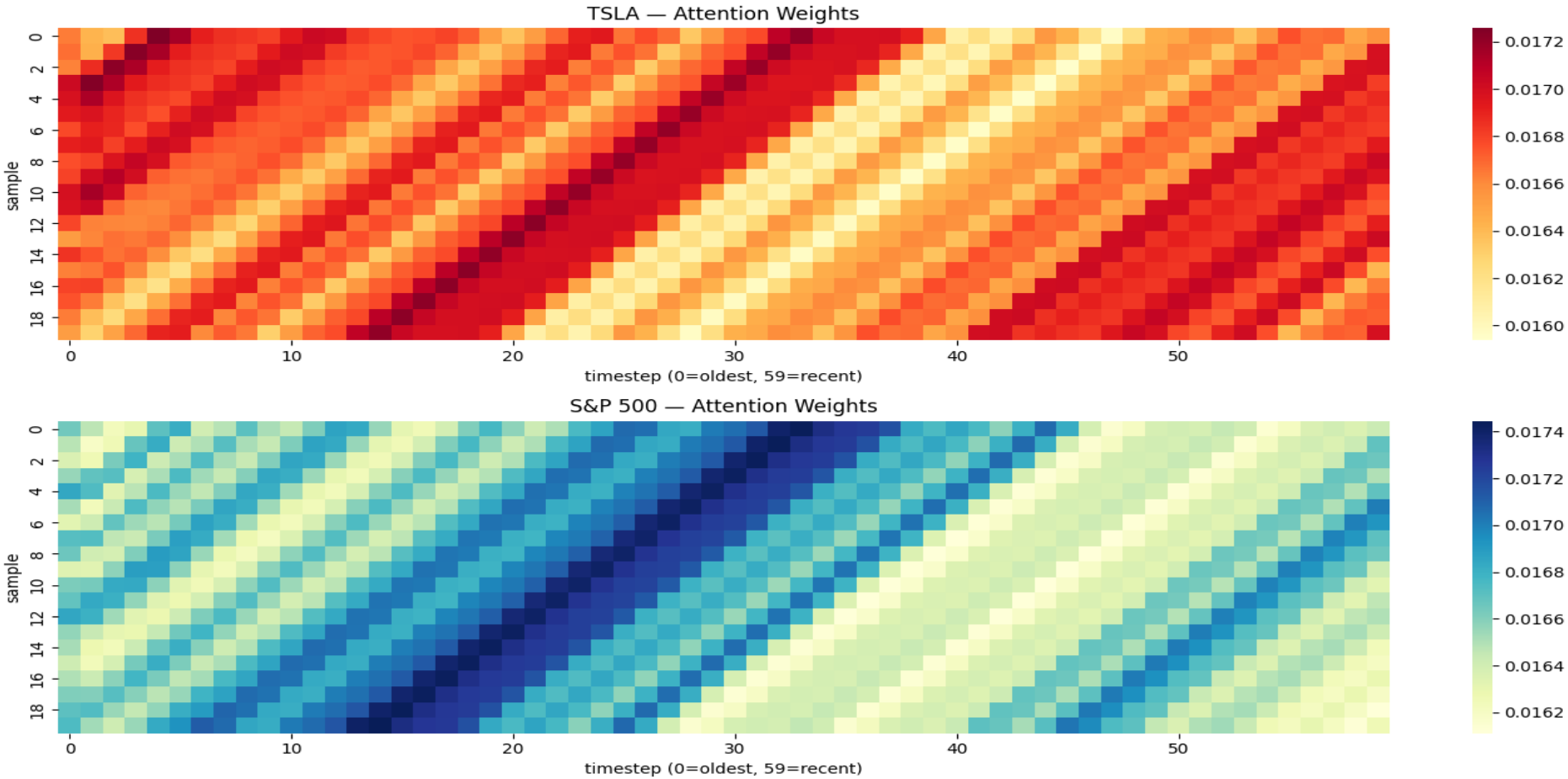
- The S&P 500 interval appears wider in absolute terms, but relative uncertainty is lower due to its larger price scale.
- TSLA shows higher relative uncertainty, indicating greater volatility captured by the TFT model.

Ensemble Predictions



The ensemble model follows the actual trend more accurately by leveraging the complementary strengths of LSTM-Attention and TFT

Attention Weights — LSTM



- Darker regions indicate higher attention weight
- Model focuses more on recent timesteps (right side)
- Confirms LSTM-Attention learns meaningful temporal dependencies

Quantitative Results — TSLA

Model	MAE	RMSE	MAPE%	DA%
LSTM-Attention	14.19	21.36	5.15	53.88
TFT	13.69	19.76	5.57	52.33
Ensemble	9.89	13.09	4.95	50.65

Table: TSLA — Model Comparison

- Ensemble achieves 38.7% lower RMSE than LSTM alone
- TFT outperforms LSTM on MAE and RMSE

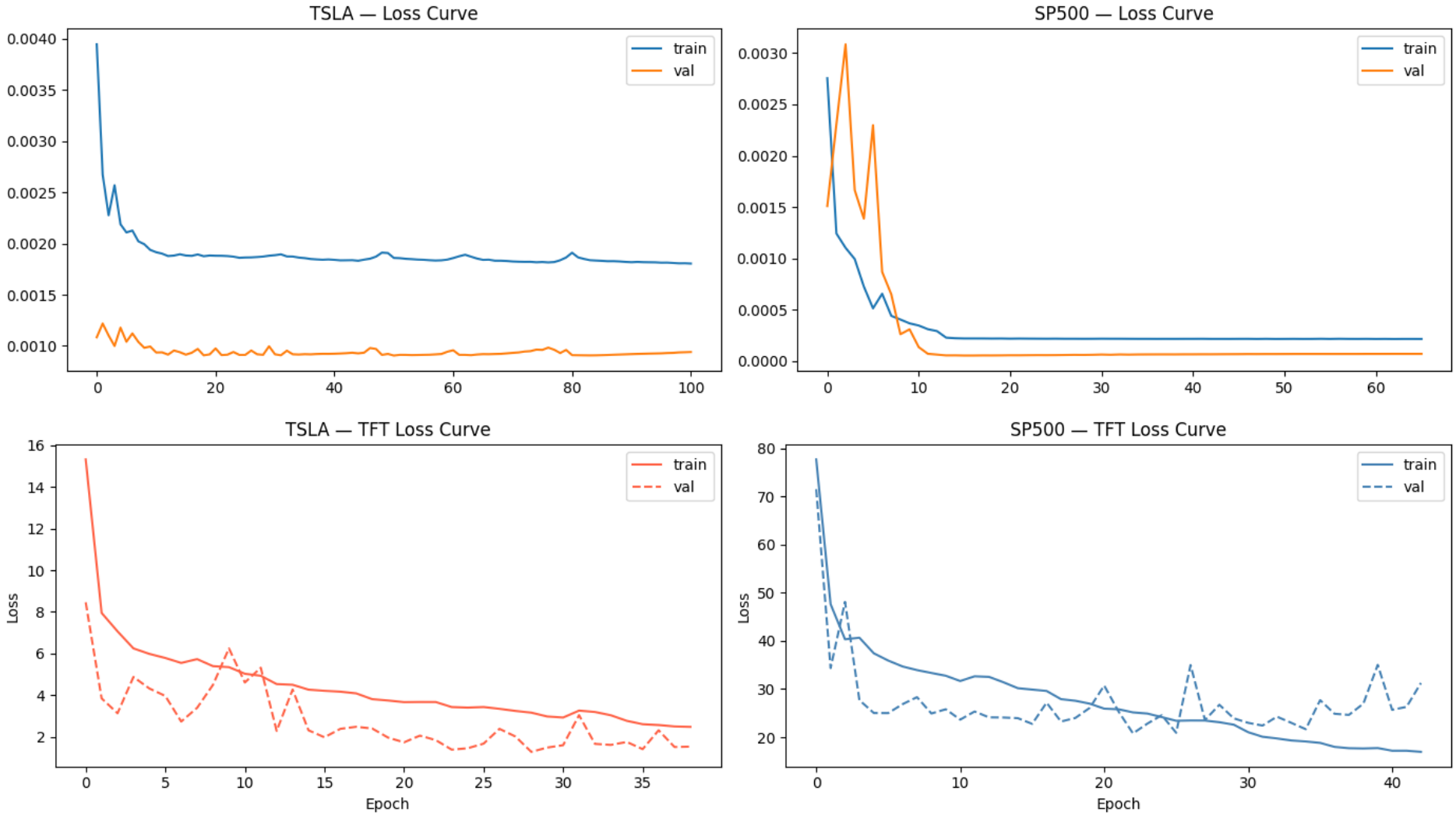
Quantitative Results — S&P 500

Model	MAE	RMSE	MAPE%	DA%
LSTM-Attention	69.56	92.27	1.23	47.29
TFT	61.26	83.22	1.11	55.77
Ensemble	62.59	82.54	1.17	54.78

Table: S&P 500 — Model Comparison

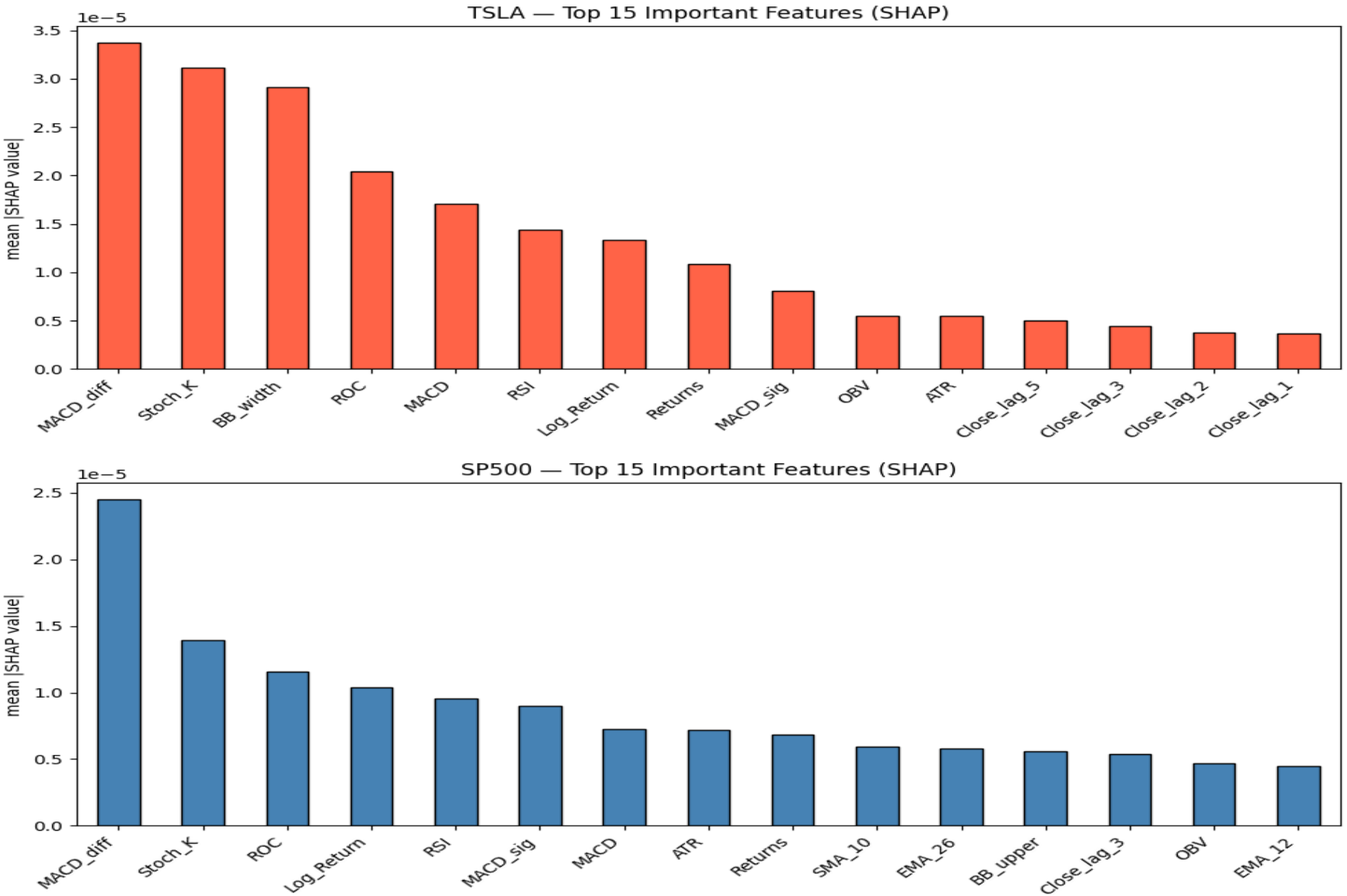
- TFT achieves best DA% (55.77%) on S&P 500
- Ensemble achieves best RMSE (82.54) on S&P 500
- Lower MAPE for S&P 500 confirms it is easier to predict than TSLA

Loss Curves



Both models converge without overfitting. Early stopping prevented over-training.

SHAP Feature Importance



SHAP Analysis — Key Findings

TSLA Top Features

1. MACD_diff
2. Stoch_K
3. BB_width
4. ROC
5. MACD

S&P 500 Top Features

1. MACD_diff
2. Stoch_K
3. ROC
4. Log_Return
5. RSI

Key Insight

MACD_diff and Stoch_K are top features for both stocks. However, BB_width (volatility) ranks 3rd for TSLA but not for S&P 500 — confirming that volatility features matter more for individual volatile stocks.

Conclusion

- **TFT outperformed LSTM-Attention** on both datasets across MAE and RMSE
- **Ensemble achieved best overall RMSE** — combining strengths of both models
- **S&P 500 is easier to predict** than TSLA (lower MAPE: 1.11% vs 4.95%)
- **SHAP revealed** momentum indicators (MACD, Stochastic) drive predictions for both stocks
- Volatility features (BB_width) are uniquely important for TSLA — not for S&P 500

Future Work

- Add sentiment analysis from financial news
- Test on more stocks across different sectors
- Explore larger transformer architectures

References

- Lim et al. (2021). *Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting*. International Journal of Forecasting.
- Bahdanau et al. (2015). *Neural Machine Translation by Jointly Learning to Align and Translate*. ICLR.
- Hochreiter & Schmidhuber (1997). *Long Short-Term Memory*. Neural Computation.
- Lundberg & Lee (2017). *A Unified Approach to Interpreting Model Predictions*. NeurIPS.
- Fischer & Krauss (2018). *Deep learning with long short-term memory networks for financial market predictions*. European Journal of Operational Research.

Thank You!